CS 2023 Main Paper

1 mark

1. -------------function is used to arrange the elements of a list in ascending order.
2. ------------- is number of tuples in a relation.
3. --------------clause is used with select statement to display data in a sorted form with respect to a specified column.

2 marks

1. Differentiate between char and varchar data types in MySQL with help of an example.
2. Name any two DDL and DML commands

3 marks

1. Write the output of the queries (i) to (iv) based on the table worker given below:

TABLE: WORKER

W_ID	F_NAME	L_NAME	CITY	STATE
102	SAHIL	KHAN	KANPUR	UTTAR PRADESH
104	SAMEER	PARIKH	ROOP NAGAR	PUNJAB
105	MARY	JONES	DELHI	DELHI
106	MAHIR	SHARMA	SONIPAT	HARYANA
107	ATHARVA	BHARDWAJ	DELHI	DELHI
108	VEDA	SHARMA	KANPUR	UTTAR PRADESH

i. SELECT F_NAME, CITY FROM WORKER ORDER BY STATE DESC;
ii. SELECT DISTINCT(CITY) FROM WORKER;
iii. SELECT F_NAME, STATE FROM WOKER WHERE L_NAME LIKE ‘’_HA%’;
iv. SELECT CITY, COUNT (*) FROM WORKER GROUP BY CITY;

2. Write the outputs of the SQL queries i) to iv) based on the relations Computers and Sales given below:

Table : COMPUTER

PROD_ID	PROD_NAME	PRICE	COMPANY	TYPE
P001	MOUSE	200	LOGITECH	INPUT
P002	LASER PRINTER	4000	CANON	OUTPUT
P003	KEYBOARD	500	LOGITECH	INPUT
P004	JOYSTICK	1000	IBALL	INPUT
P005	SPEAKER	1200	CREATIVE	OUTPUT
P006	DESKJET PRINTER	4300	CANON	OUTPUT

Table : SALES

PROD_ID	QTY_SOLD	QUARTER
P002	4	1
P003	2	2
P001	3	2
P004	2	1

i) SELECT MIN(PRICE), MAX(PRICE) FROM COMPUTERS;

ii) SELECT COMPANY, COUNT (*) FROM CUSTOMER GROUP BY COMPANY HAVING COUNT(COMPANY)>1;

iii) SELECT PROD_NAME, QTY_SOLD FROM CUSTOMER C, SALES S WHERE C.PROD_ID=S.PROD_ID AND TYPE='INPUT';

iv) SELECT PROD_NAME, COMPANY, QUARTERS FROM CUSTOMER C, SALES S WHERE C.PROD_ID=S.PROD_ID;

v) Write command to view all databases.

5 marks

The school has asked their estate manager Mr. Rahul to maintain the data of all the labs in a table LAB. Rahul has created and entered data of 5 labs.

LABNO	LAB_NAME	INCHARGE	CAPACITY	FLOOR
L001	CHEMISTRY	Daisy	20	I
L002	BIOLOGY	Venky	20	II
L003	MATH	Preeti	15	I
L004	LANGUAGE	Daisy	36	III
L005	COMPUTER	Mary Kom	37	II

Based on the data given above answer the following questions:
1. Identify the columns which can be considered as candidate keys.
2. Write the degree and cardinality of the table.
3. Write the statement to:
   a) Insert a new row with appropriate data.
   b) Increase the capacity of all the labs by 10 students which are on ‘I’ floor.

CS Sample Paper 2023

1 mark

1. --------- command is used to remove primary key from the table in SQL.
2. Which of the following commands will delete the table from MYSQL database?
   a) DELETE TABLE
   b) DROP TABLE
   c) REMOVE TABLE
   d) ALTER TABLE
3. ---------- is a non-key attribute, whose values are derived from the primary key of some other table.
   a) Primary Key
   b) Foreign Key
   c) Candidate Key
   d) Alternate Key
4. The SELECT statement when combined with --------------clause, returns records without repetition.
5. Which function is used to display the total number of records from table in a database?
   a) sum(*)
   b) total(*)
   c) count(*)
   d) return(*)

2 marks

1. Explain the use of 'Foreign Key' in a Relational Database Management System. Give example to support your answer.
2. Differentiate between count () and count (*) functions in SQL with appropriate example.

OR

Categorize the following commands as DDL or DML: INSERT, UPDATE, ALTER, DROP

3 marks

1. a) Consider the following tables – Bank_Account and Branch:

Table: Bank_Account

ACode	Name	Type
A01	Amrita	Savings
A02	Parthodas	Current
A03	Miraben	Current

Table: Branch

ACode	City
A01	Delhi
A02	Mumbai
A01	Nagpur

What will be the output of the following statement?
SELECT * FROM Bank_Account NATURAL JOIN Branch;
b) Write the output of the queries (i) to (iv) based on the table, TECH_COURSE given below:

Table: TECH_COURSE

CID	CNAME	FEES	STARTDATE	TID
C201	Animation and VFX	12000	2022-07-02	101
C202	CADD	15000	2021-11-15	NULL
C203	DCA	10000	2020-10-01	102
C204	DDTP	9000	2021-09-15	104
C205	Mobile Application Development	18000	2022-11-01	101
C206	Digital marketing	16000	2022-07-25	103

i. SELECT DISTINCT TID FROM TECH_COURSE;
ii. SELECT TID, COUNT(*), MIN(FEES) FROM TECH_COURSE GROUP BY TID HAVING COUNT(TID)>1;
iii. SELECT CNAME FROM TECH_COURSE WHERE FEES>15000 ORDER BY CNAME;
iv. SELECT AVG(FEES) FROM TECH_COURSE WHERE FEES BETWEEN 15000 AND 17000;

2. (a) Write the outputs of the SQL queries (i) to (iv) based on the relations Teacher and Placement given below:

Table : Teacher

T_ID	Name	Age	Department	Date_of_join	Salary	Gender
1	Arunan	34	Computer Sc	2019-01-10	12000	M
2	Saman	31	History	2017-03-24	20000	F
3	Randeep	32	Mathematics	2020-12-12	30000	M
4	Samira	35	History	2018-07-01	40000	F
5	Raman	42	Mathematics	2021-09-05	25000	M
6	Shyam	50	History	2019-06-27	30000	M
7	Shiv	44	Computer Sc	2019-02-25	21000	M
8	Shalakha	33	Mathematics	2018-07-31	20000	F

Table : Placement

P_ID	Department	Place
1	History	Ahmedabad
2	Mathematics	Jaipur
3	Computer Sc	Nagpur

i. SELECT Department, avg(salary) FROM Teacher GROUP BY Department;
ii. SELECT MAX(Date_of_Join), MIN(Date_of_Join) FROM Teacher;
iii. SELECT Name, Salary, T.Department, Place FROM Teacher T, Placement P WHERE T.Department = P.Department AND Salary>20000;
iv. SELECT Name, Place FROM Teacher T, Placement P 3 8 WHERE Gender ='F' AND T.Department=P.Department;

b) Write the command to view all tables in a database.

5 marks

Navdeep creates a table RESULT with a set of records to maintain the marks secured by students in Sem 1, Sem2, Sem3 and their division. After creation of the table, he has entered data of 7 students in the table.

ROLL_NO	SNAME	SEM1	SEM2	SEM3	DIVISION
101	KARAN	366	410	402	I
102	NAMAN	300	350	325	I
103	ISHA	400	410	415	I
104	RENU	350	357	415	I
105	ARPIT	100	75	178	IV
106	SABINA	100	205	217	II
107	NEELAM	470	450	471	I

Based on the data given above answer the following questions:

i. Identify the most appropriate column, which can be considered as Primary key.

ii. If two columns are added and 2 rows are deleted from the table result, what will be the new degree and cardinality of the above table?

iii. Write the statements to: a. Insert the following record into the table Roll No- 108, Name- Aadit, Sem1- 470, Sem2-444, Sem3 475, Div – I.

iv. Increase the SEM2 marks of the students by 3% whose name begins with 'N'.

OR (Option for part iv only)

Write the statements to: a. Delete the record of students securing IV division. b. Add a column REMARKS in the table with datatype as varchar with 50 characters

CS 2023 Sahodya Paper

1 Mark

1. An attribute in a relation is a foreign key if it is the ____________ key in any other relation.
   a) candidate b) primary
   c) alternate d) foreign
2. The NULL value also means ____________.
   a) value equal to zero
   b) unknown value
   c) negative values
   d) a large value

3. Which clause is used to sort the rows of final result set by one or more columns?
   a) HAVING b) ORDER BY
   c) WHERE d) FROM
4. The referential integrity constraint of a relational database can be specified with the help of ________.

2 Marks

1. What is data redundancy? List any two problems associated with it.
2. What do you understand by the term primary key and degree of a relation in relational database? Illustrate with the help of an example.

3 Marks

1. Consider the following two tables:

Table: Items

I-ID	Item-Name	Manufacturer	Price
PC01	Personal Computer	ABC	35000
LC05	Laptop	ABC	55000
PC03	Personal Computer	XYZ	32000
PC06	Personal Computer	COM	37000
LC03	Laptop	PQR	57000

Table: Customer

C-Id	Customer-Name	City	I_Id
01	N Roy	Delhi	LC03
06	H Singh	Mumbai	PC03
12	R Pandey	Delhi	PC06
15	C Sharma	Delhi	LC03
16	K Agarwal	Bangalore	PC01

a) What will the output of the following statement be?
   SELECT Customer_Name, City, Item_Name, Price FROM Customer, Item WHERE Customer.I_Id==Items.I_ID;

b) Write the output of the queries (i) to (iv) based on the table, Books given below:

No	Title	Author	Type	Pub	Qty	Price
1	Data Structure	Lipschutz	DS	McGraw	4	217
2	Computer Studies	French	FND	Galgotia	2	75
3	Advanced Pascal	Schildt	PROG	McGraw	4	350
4	Dbase Dummies	Palmer	DBMS	Pustak	5	130
5	Mastering C++	Gruewich	PROG	BPB	3	295
6	Guide Network	Freed	NET	Zpress	3	200
7	Mastering Foxpro	Seigal	DBMS	BPB	2	135
8	DOS Guide	Norton	OS	PHI	3	175
9	Basic for Beginners	Norton	PROG	BPB	3	40
10	Mastering Windows	Cowart	OS	BPB	1	225

i) SELECT Title, Author, Pub FROM Books WHERE Type LIKE 'OS';

ii) SELECT DISTINCT(Pub) FROM Books WHERE Qty>3;

iii) SELECT SUM(Price), Type FROM Books GROUP BY Pub HAVING Pub LIKE 'BPB';

iv) SELECT AVG(Price) FROM Books WHERE Qty BETWEEN 1 AND 2;

2. a) Write the outputs of the SQL Queries based on the relations given below:

Table: School

Code	TeacherName	Subject	DOJ	Classes	Experience
1001	Uma Shankar	English	12/03/2000	24	10
1009	Nandita Rai	Physics	03/09/1998	26	12
1203	Lisa Anand	English	09/04/2000	27	5
1045	Yashraj	Maths	24/08/2000	24	15
1123	Jeevan	Physics	16/07/1999	28	4
1167	Harish B	Chemistry	19/10/1999	27	5
1215	Ramesh	Physics	11/05/1998	22	16

Table: Admin

Code	Gender	Designation
1001	Male	Vice Principal
1009	Female	Coordinator
1203	Female	Coordinator
1045	Male	HOD
1123	Male	Senior Teacher
1167	Male	Senior Teacher
1215	Male	HOD

i) SELECT TeacherName, Classes FROM School WHERE Classes>25 ORDER BY TeacherName;

ii) SELECT TeacherName, Admin.Code, Gender FROM School, Admin WHERE School.Code=Admin.Code AND Gender LIKE 'Female';

iii) SELECT TeacherNAme, Designation FROM School, Admin WHERE School.Code=Admin.Code AND Subject LIKE 'English';

iv) SELECT Subject, Classes FROM School GROUP BY Subject HAVING MIN(Classes);

3 Marks

Sahana is creating a table named football as shown below in the database sports.

Team: Football

Team-ID	**Team-Name**	**Sponsor-Id**	**Price**
TA01	Chelsea	S01	120000
TA02	Sunderland	S03	50000
TB03	Fulham	S07	67000
TE04	Blackpool	S09	12000
TL05	Blackburn Rovers	S03	135000
TM06	Manchester City	S21	2000000
TT07	Bolton Wanderers	S25	5000
TT08	Wigan Athletic	S03	25000

Based on the above table answer the following questions:

a) What will the cardinality of the table football?

b) Which key do you identify as a primary key?

c) Write statements to:

   i) To add a column named Team_Colour of 10 characters to the table.

   ii) To delete tuple with price 50000 from the table.

OR

   i) To increase the price of blackpool team to 15000.

   ii) To insert another record as follows: (TT08, Wigan Athletic, S03, 2500)

CS Supplementary 2023

1 Mark

1. In a relational model, tables are called------------, that store data for different columns.
   (a) Attributes
   (b) Degrees
   (c) Relations

2. ---------------statement of SQL is used to insert new records in a table.
(a) ALTER
(b) UPDATE
(c) INSERT
(d) CREATE

2 Marks

1. Explain the usage of HAVING clause in GROUP BY command in RDBMS with the help of an example.
2. Differentiate between IN and BETWEEN operators in SQL with appropriate examples.

OR

Which of the following is NOT a DML command?
DELETE, DROP, INSERT, UPDATE

3 Marks

1. a) Consider the following tables Student and Sport :

Table : Student

ADMNO	NAME	CLASS
1100	MEENA	X
1101	VANI	XI

Table : Sport

ADMNO	GAME
1100	CRICKET
1103	FOOTBALL

What will be the output of the following statement?
SELECT * FROM Student, Sport;
(b) Write the output of the queries (i) to (iv) based on the table, GARMENT given below :

TABLE : GARMENT

GCODE	TYPE	PRICE	FCODE	ODR_DATE
G101	EVENING GOWN	850	F03	2008-12-19
G102	SLACKS	750	F02	2020-10-20
G103	FROCK	1000	F01	2021-09-09
G104	TULIP SKIRT	1550	F01	2021-08-10
G105	BABY TOP	1500	F02	2020-03-31
G106	FORMAL PANT	1250	F01	2019-01-06

i. SELECT DISTINCT(COUNT(FCODE))FROM GARMENT;
ii. SELECT FCODE, COUNT(*), MIN(PRICE) FROM GARMENT GROUP BY FCODE HAVING COUNT(*)>1;
iii. SELECT TYPE FROM GARMENT WHERE ODR_DATE >'2021-02-01' AND PRICE < 1500;
iv. SELECT * FROM GARMENT WHERE TYPE LIKE ‘F%’;

2. Write the output of any three SQL queries (i) to (iv) based on the tables COMPANY and CUSTOMER given below:

Table : COMPANY

CID	C_NAME	CITY	PRODUCTNAME
111	SONY	DELHI	TV
222	NOKIA	MUMBAI	MOBILE
333	ONIDA	DELHI	TV
444	SONY	MUMBAI	MOBILE
555	BLACKBERRY	CHENNAI	MOBILE
666	DELL	DELHI	LAPTOP

Table : CUSTOMER

CUSTID	CID	NAME	PRICE	QTY
C01	222	ROHIT SHARMA	70000	20
C02	666	DEEPIKA KUMARI	50000	10
C03	111	MOHAN KUMAR	30000	5
C04	555	RADHA MOHAN	30000	11

i. SELECT PRODUCTNAME, COUNT(*)FROM COMPANY GROUP BY PRODUCTNAME HAVING COUNT(*)> 2;

ii. SELECT NAME, PRICE, PRODUCTNAME FROM COMPANY C, CUSTOMER CT WHERE C.CID = CU.CID AND C_NAME = 'SONY';

iii. SELECT DISTINCT CITY FROM COMPANY; (iv) SELECT * FROM COMPANY WHERE C_NAME LIKE '%ON%';

5 Marks

The ABC Company is considering to maintain their salespersons records using SQL to store data. As a database administrator, Alia created the table Salesperson and also entered the data

Table : Salesperson

S_ID	S_NAME	AGE	S_AMOUNT	REGION
S001	SHYAM	35	20000	NORTH
S002	RISHABH	30	25000	EAST
S003	SUNIL	29	21000	NORTH
S004	RAHIL	39	22000	WEST
S005	AMIT	40	23000	EAST

Based on the data given above, answer the following questions:

i) Identify the attribute that is best suited to be the Primary Key and why?

ii) The Company has asked Alia to add another attribute in the table. What will be the new degree and cardinality of the above table?

iii) Write the statements to:

(a) Insert details of one salesman with appropriate data.

(b) SHYAM SOUTH table Salesperson.

OR (Option for part iii only)

Write the statement to:

(a) Delete the record of salesman RISHABH, as he has left the company.

(b) Remove an attribute REGION from the table.

CS MAIN PAPER 2024

1 Mark

i) The SELECT statement when combined with ----------clause, returns records without repetition.

ii) In SQL, the aggregate function which will display the cardinality of the table is --------_.

2 Marks

a) Ms. Veda created a table named Sports in a MySQL database, containing columns Game-id, P-age and G-name. after creating the table, she realized that the attribute, category has to be added. Help her to write a command to add the category column, Thereafter, write the command to insert the following record in the table: Game Id: G42, P-age above 18, G-name: Chess and Cateogory: Senior.

b) Write the SQL commands to perform the following tasks:

i) View the list of tables in the database, exam.

ii) View the structure of table, Term1.

3 Marks:

1. Consider the table ORDERS given below and write the output of SQL queries that follows:

ORDNO	ITEM	QTY	RATE	ORDATE
1001	RICE	23	120	2023-09-10
1002	PULSES	13	120	2023-10-18
1003	RICE	25	110	2023-11-17
1004	WHEAT	28	65	2023-12-25
1005	PULSES	16	110	2024-01-15
1006	WHEAT	27	55	2024-04-15
1007	WHEAT	25	60	2024-04-30

i) SELECT ITEM, SUM(QTY) FROM RDERS GROUP BY ITEM;

ii) SELECT ITEM, QTY FROM ORDERS WHERE ORDATE BETWEEN ‘2023-11-01’ AND ‘2023-12-31’;

iii) SELECT ORDNO, ORDATE FROM ORDERS WHERE ITEM=’WHEAT’ AND RATE >=60;

2. Consider the table Projects given below:

Table : Projects

P_id	Pname	Language	Startdate	Enddate
P001	School Management System	Python	2023-01-12	2023-04-03
P002	Hotel Management System	C++	2022-12-01	2023-02-02
P003	Blood Bank	Python	2023-02-11	2023-03-02
P004	Payroll Management System	Python	2023-03-12	2023-06-02

Based on the given table, write SQL queries for the following:

i) Add the constraint, primary key to column P-Id in the existing table.

ii) To change the language to Python of the project whose P-Id is P002

iii) To delete the table Projects from MySQL database along with the data.

4 Marks

Consider the tables admin and transport given below:

Table : Admin

S_id	S_name	Address	S_type
S001	Sandhya	Rohini	Day Boarder
S002	Vedanshi	Rohtak	Day Scholar
S003	Vibhu	Raj Nagar	NULL
S004	Atharva	Rampur	Day Boarder

Table : Transport

S_id	Bus_no	Stop_name
S002	TSS10	Sarai Kale Khan
S004	TSS12	Sainik Vihar
S005	TSS10	Kamla Nagar

Write SQL Queries for the following:

i) Display the student's name and their stop name from the tables admin and transport.

ii) Display the number of students whose S-type is not known.

iii) Display all details of the students whose name starts with 'v'.

iv) Display student id and address in alphabetical order of student name from the table admin.

CS SAMPLE PAPER 2024

1 Mark

1. In MYSQL database, if a table, Alpha has degree 5 and cardinality 3, and another table, Beta has degree 3 and cardinality 5, what will be the degree and cardinality of the Cartesian product of Alpha and Beta?
   a) 5,3
   b) 8,15
   c) 3,5
   d) 15,8
2. Which of the following statements is FALSE about keys in a relational database?
   a) Any candidate key is eligible to become a primary key.
   b) A primary key uniquely identifies the tuples in a relation.
   c) A candidate key that is not a primary key is a foreign key.
   d) A foreign key is an attribute whose value is derived from the primary key of another relation.

2 Marks

Ms. Shalini has just created a table named "Employee" containing columns Ename, Department and Salary. After creating the table, she realized that she has forgotten to add a primary key column in the table. Help her in writing an SQL command to add a primary key column EmpId of integer type to the table Employee.
Thereafter, write the command to insert the following record in the table:
EmpId- 999 Ename- Shweta Department: Production Salary: 26900

OR

Zack is working in a database named SPORT, in which he has created a table named "Sports" containing columns SportId, SportName, no_of_players, and category. After creating the table, he realized that the attribute, category has to be deleted from the table and a new attribute TypeSport of data type string has to be added. This attribute TypeSport cannot be left blank. Help Zack write the commands to complete both the tasks.

3 Marks

1. Consider the table CLUB given below and write the output of the SQL queries that follow.

CID	CNAME	AGE	GENDER	SPORTS	PAY	DOAPP
5246	AMRITA	35	DEMALE	CHESS	900	2006-03-27
4687	SHYAM	37	MALE	CRICKET	1300	2004-04-15
1245	MEENA	23	FEMALE	VOLLEYBALL	1000	2007-06-18
1622	AMRIT	28	MALE	KARATE	1000	2007-09-05
1256	AMINA	36	FEMALE	CHESS	1100	2003-08-15
1720	MANJU	33	FEMALE	KARATE	1250	2004-04-10
2321	VIRAT	35	MALE	CRICKET	1050	2005-04-30

i) SELECT COUNT(DISTINCT SPORTS) FROM CLUB;
ii) SELECT CNAME, SPORTS FROM CLUB WHERE DOAPP<"2006-04-30" AND CNAME LIKE "%NA";
iii) SELECT CNAME, AGE, PAY FROM CLUB WHERE GENDER = "MALE" AND PAY BETWEEN 1000 AND 1200;

2. Consider the table Personal given below:

P_ID	NAME	DESIG	SALARY	ALLOWANCE
P01	ROHIT	MANAGER	89000	4800
P02	KASHISH	CLERK	NULL	1600
P03	MAHESH	SUPERVISER	48000	NULL
P04	SALIL	CLERK	31000	1900
P05	RAVINA	SUPERVISOR	NULL	2100

Based on the given table, write SQL queries for the following:
i) Increase the salary by 5% of personals whose allowance is known.
ii) Display Name and Total Salary (sum of Salary and Allowance) of all personals. The column heading 'Total Salary' should also be displayed.
iii) Delete the record of personals who have salary greater than 25000

4 Marks

Consider the tables PRODUCT and BRAND given below:

Table: Product

PCODE	PNAME	UPRICE	RATING	BID
P01	Shampoo	120	6	M03
P02	Toothpaste	54	8	M02
P03	Soap	25	7	M03
P04	Toothpaste	65	4	M04
P05	Soap	38	5	M05
P06	Shampoo	245	6	M05

Table: Brand

BID	BNAME
M02	Dant Kanti
M03	Medimix
M04	Pepsodent
M05	Dove

Write SQL queries for the following:
i) Display product name and brand name from the tables PRODUCT and BRAND.
ii) Display the structure of the table PRODUCT.
iii) Display the average rating of Medimix and Dove brands
iv) Display the name, price, and rating of products in descending order of rating.

CS 2024 Sahodya Paper

1 Marks

1. Select the DDL command from the following:
   a) SELECT b) INSERT c) DELETE d) DROP

2. Which of the following will display information about all the employees in the emp table, whose names contains second letter as ‘S’?
   a) SELECT * FROM emp WHERE name like ‘_S%’;
   b) SELECT * FROM emp WHERE name like ‘%S_’;
   c) SELECT * FROM emp WHERE name like ‘__S%’;
   d) SELECT * FROM emp WHERE name like ‘S%’;
3. Vamshi creates many tables under database ‘Exam’, now she wants to list out all the tables created in that. Write a SQL query to execute the above condition.
4. The update clause in MySQL can:
   a) Update only one row at a time
   b) Update more than one row at a time
   c) Delete more than one row at a time
   d) Delete only one row at a time

2 Marks

Given two tables:
Student (Rollno, Admno, Name, Class) and Fees (FeeID, Rollno, Amount)
Develop a SQL query to combine two table with the help of foreign key.

OR

Ruby is working on a database named ‘Client’, in which she created a table name ‘Order’ containing columns Order_ID, Order_Name, Amount and DOD. After creating the table, she noticed that the customer’s name is not mentioned. So, she decided to add a new column ‘C_Name’. She also decided that the category DOD is not required and has to be deleted. Help her to write the commands to complete the task.

3 Marks

1. A vegetable store ‘Fresh Stock’ maintains the inventory using a SQL table. The table is given below:

ItemCode	ItemName	SCode	Qty
1001	Potato	11	50
1002	Onion	11	50
1003	Tomato	12	25
1004	Cauliflower	13	50
1005	Garlic	14	20
1006	Ginger	14	25

   i) Name the columns that are suitable for candidate keys.
   ii) Perform insert query: (107, Ginger, 15)
   iii) Write a query to add a column price to the existing table.

2. Write the outputs of the SQL Queries given below based on the relation Teacher given below:

T-ID	Name	Age	Department	DOJ	Salary	Gender
1	Jugal	34	Comp Sc	10/01/2017	12000	M
2	Sharmila	31	History	24/03/2008	20000	F
3	Sandeep	32	Mathematics	12/12/2016	30000	M
4	Sangeeta	35	History	01/07/2015	40000	F
5	Rakesh	42	Mathematics	05/09/2007	25000	M
6	Shyam	50	History	27/06/2008	30000	M
7	Shiv Om	44	Comp Sc	25/02/2017	21000	M
8	Shalakha	33	Mathematics	31/07/2018	20000	F

   i) SELECT Name, Age, Department FROM Teacher WHERE Department='History' or gender='F';
   ii) SELECT * FROM Teacher WHERE Age>=40;
   iii) SELECT Name, Department, Salary WHERE Department NOT IN ('Mathematics');

4 Marks

Write the output of the queries (i) to (iv) based on the table, Tech-Course given below:

CID	Cname	Fees	StartDate	TID
C201	Animation and VFX	12000	2022-07-02	101
C202	CADD	15000	2021-11-15	NULL
C203	DCA	10000	2020-10-01	102
C204	DDTP	9000	2021-09-15	104
C205	Mobile Application	18000	2022-11-01	101
C206	Digital Marketing	16000	2022-07-25	103

   i) SELECT DISTINCT TID FROM Tech-Course;
   ii) SELECT TID, COUNT(*), MIN(Fees) FROM Tech-Course GROUP BY TID HAVING COUNT(TID)>1;
   iii) SELECT Cname, FROM Tech-Course WHERE Fees>15000 ORDER BY Cname;
   iv) SELECT Avg(Fees) FROM Tech-Course WHERE BETWEEN 15000 AND 17000;

CS SUPPLEMENTARY PAPER 2024

1 Marks

1. In SQL, which command will be used to add a new record in a table?
   a) UPDATE
   b) ADD
   c) INSERT
   d) ALTER TABLE
2. Mr. Ravi is creating a field that contains alphanumeric values and fixed lengths. Which MySQL data type should he choose for the same?
   a) VARCHAR
   b) CHAR
   c) LONG
   d) NUMBER

2 Marks

Mr. Atharva is given a task to create a database, Admin. He has to create a table, users in the database with the following columns:
User-id – int
User-name – varchar(20)
Password – varchar (10)
Help him by writing SQL queries for both tasks.

OR

Ms. Rita is a database administrator at a school. She is working on the table, student containing the columns like Stud-id, Name, Class and Stream. She has been asked by the principal to strike off the record of a student named Rahul with student-id as 100 from the school records and add another student who has been admitted with the following details:
Stud-id – 123 Name – Rajeev Class – 12 Stream – Science
Help her by writing SQL queries for both tasks.

3 Marks

1. Consider the table Stationery given below and write the output of the SQL queries that follow.

Table: Stationery

ITEMNO	ITEM	DISTRIBUTOR	QTY	PRICE
401	Ball Pen 0.5	Reliable Stationers	100	16
402	Gel Pen Premium	Classic Plastics	150	20
403	Eraser Big	Clear Deals	210	10
404	Eraser Small	Clear Deals	200	5
405	Sharpener Classic	Classic Plastics	150	8
406	Gel Pen Classic	Classic Plastics	100	15

i) SELECT DISTRIBUTOR, SUM(QTY) FROM STATIONERY GROUP BY DISTRIBUTOR;
ii) SELECT ITEMNO, ITEM FROM STATIONERY WHERE DISTRIBUTOR = "Classic Plastics" AND PRICE > 10;
iii) SELCET ITEM, QTY * PRICE AS "AMOUNT" FROM STATIONERY WHERE ITEMNO = 402;

2. Consider the table Rent_cab, given below:

Table : Rent_cab

Vcode	VName	Make	Color	Charges
101	Big car	Carus	White	15
102	Small car	Polestar	Silver	10
103	Family car	Windspeed	Black	20
104	Classic	Studio	White	30
105	Luxury	Trona	Red	9

Based on the given table, write SQL queries for the following:

i) Add a primary key to a column name Vcode.

ii) Increase the charges of all the cabs by 10%.

iii) Delete all the cabs whose maker name is "Carus".

4 Marks

Consider the tables GAMES and PLAYERS given below:

Table: GAMES

GCode	GameName	Type	Number	PrizeMoney
101	Carrom Board	Indoor	2	5000
102	Badminton	Outdoor	2	12000
103	Table Tennis	Indoor	4	NULL
104	Chess	Indoor	2	9000
105	Lawn Tennis	Outdoor	4	25000

Table: PLAYERS

PCode	Name	GCode
1	Nabi Ahmad	101
2	Ravi Sahai	108
3	Jatin	101
4	Nazneen	103

Write SQL queries for the following:

i) Display the game type and average number of games played in each type.

ii) Display prize money, name of the game, and name of the players from the tables Games and Players.

iii) Display the types of games without repetition.

iv) Display the name of the game and prize money of those games whose prize money is known.

CS MAIN PAPER 2025

1 Marks

1. While creating a table, which constraint does not allow insertion of duplicate values in a table?
2. Which of the following is a DML command in SQL?
   a) UPDATE
   b) CREATE
   c) ALTER
   d) DROP

3. Which aggregate function in SQL, displays the number of values in the specified column ignoring the NULL values?
   a) len()
   b) count()
   c) number()
   d) num()
4. In MySQL, which type of value should not be enclosed within quotation marks?
   a) date
   b) varchar
   c) float
   d) char
5. Sate True or false:
   If table A has 6 rows and 3 columns, and table B has 5 rows and 2 columns, the cartesian product of A and B have 30 rows and 5 columns.

2 Marks

Nisha is assigned the task of maintaining the staff data of an organization. She has to store details of the staff in the SQL table named Employees with attributes- Empno, Name, Department and Basicsal to store employee's identification number, name, department and basic salary respectively. There can be two or more employees with the same name in the organization.

i) a) Help Nisha to identify the attribute which should be designated as Primary Key. Justify your answer.

OR

b) Help Nisha to identify the constraint which should be applied to the attribute Name such that the employee's name cannot be left empty or NULL while entering the records but can have duplicate values.

ii) a) Write the SQL command to change the size of the attribute basicsal in the table employees to allow maximum value of 99999.99 to be stored in it

OR

Write the SQL command to delete the table employees.

4 Marks

1. Suman has created a table named WORKER with a set of records to maintain the data of the construction sites, which consists of WID, WNAME, WAGE, HOURS, TYPE and SITEID. After creating the table, she entered data in it, which is as follows:

WID	WNAME	WAGE	HOURS	TYPE	SITEID
W01	Ahmed J	1500	200	Unskilled	103
W11	Naveen S	520	100	Skilled	101
W02	Jacob B	780	95	Unskilled	101
W15	Nihal K	560	110	Semiskilled	NULL
W10	Anju S	1200	130	Skilled	103

a) Based on the data given above, answer the following questions:
   i) Write the SQL statement to display the names and wages of those workers whose wages are between 800 and 1500.
   ii) Write the SQL statement to display the records of the workers whose SITEID is not known.
   iii) Write the SQL statement to display WNAME, WAGE and HOURS of all those workers whose TYPE is skilled.
   iv) Write the SQL statement to change the WAGE to 1200of the workers where the TYPE is Semiskilled.

OR

b) Considering the above given table WORKER, write the output on the execution of the following SQL statements:
   i) SELECT WNAME, WAGE*HOURS FROM WORKER WHERE SITEID=103;
   ii) SELECT COUNT(DISTINCT(TYPE)) FROM WORKER;
   iii) SELECT MAX(WAGE), MIN(WAGE), TYPE FORM WORKER GROUP BY TYPE;
   iv) SELECT WNAME, SITEID FROM WORKER WHERE TYPE='Unskilled' ORDER BY HOURS;

2. Assume that you are working in the IT Department of a Creative Art Gallery (CAG), which sells different forms of art creations like paintings, sculptures etc. The data of Art Creations and Artists are kept in tables Articles and Artists respectively. Following are few records form these tables:

Table: Articles

Code	A_Code	Article	DOC	Price
PL001	A0001	Painting	2018-10-19	20000
SC028	A0004	Sculpture	2021-01-15	16000
QL005	A0003	Quilling	2024-04-24	3000

Table: Artists

A_Code	Name	Phone	Email	DOB
A0001	Roy	595923	r@CrAG.com	1986-10-12
A0002	Ghosh	1122334	ghosh@CrAG.com	1972-02-05
A0003	Gargi	121212	Gargi@CrAG.com	1996-03-22
A0004	Mustafa	33333333	Mf@CrAg.com	2000-01-01

Note: The tables contain many more records than shown here. DOC is the date of creation of an article.

As an employee of CAG, you are required to write the SQL queries for the following:

i) To display all the records from the Articles table in the descending order of the price.
ii) To display the details of articles which are created in the year 2020.
iii) To display structure of Artists table.
iv) To display the name of all artists whose article is painting through Equi Join.

OR

To display the name of all artists whose article is painting through Natural Join.

CS SAMPLE PAPER 2025

1 Marks

1. If a table which has one Primary key and two alternate keys. How many Candidate keys will this table have?
   a) 1
   b) 2
   c) 3
   d) 4
2. Which SQL command can change the degree of an existing relation?

3. What will be the output of the query? SELECT * FROM products WHERE product-name LIKE 'App%';
   a) Details of all products whose names start with 'App'
   b) Details of all products whose names end with 'App'
   c) Names of all products whose names start with 'App'
   d) Names of all products whose names end with 'App'
4. In which datatype the value stored is padded with spaces to fit the specified length.
   a) DATE
   b) VARCHAR
   c) FLOAT
   d) CHAR
5. Which aggregate function can be used to find the cardinality of a table?
   a) sum() c) avg()
   b) count() d) max()

2 Marks

1. A) What constraint should be applied on a table column so that duplicate values are not allowed in that column, but NULL is allowed.

OR

B) What constraint should be applied on a table column so that NULL is not allowed in that column, but duplicate values are allowed.

2. A) Write an SQL command to remove the Primary Key constraint from a table, named MOBILE. M_ID is the primary key of the table.

OR

B) Write an SQL command to make the column M_ID the Primary Key of an already existing table, named MOBILE.

4 Marks

1. Consider the table ORDERS as given below:

O_Id	C_Name	Product	Quantity	Price
1001	Jitendra	Laptop	1	12000
1002	Mustafa	Smartphone	2	10000
1003	Dhwani	Headphone	1	1500

Note: The table contains many more records than shown here.

a) Write the following queries:
   i) To display the total Quantity for each Product, excluding Products with total Quantity less than 5.
   ii) To display the orders table sorted by total price in descending order.
   iii) To display the distinct customer names from the Orders table.
   iv) Display the sum of Price of all the orders for which the quantity is null.

OR

b) Write the output:
   i) Select c-name, sum(quantity) as total-quantity from orders group by c-name;
   ii) Select * from orders where product like '%phone%';
   iii) Select o-id, c-name, product, quantity, price from orders where price between 1500 and 12000;
   iv) Select max(price) from orders;

2. Saman has been entrusted with the management of Law University Database. He needs to access some information from FACULTY and COURSES tables for a survey analysis. Help him extract the following information by writing the desired SQL queries as mentioned below.

Table: FACULTY

F_ID	FName	LName	Hire_Date	Salary
102	Amit	Mishra	12-10-1998	12000
103	Nitin	Vyas	24-12-1994	8000
104	Rakshit	Soni	18-5-2001	14000
105	Rashmi	Malhotra	11-9-2004	11000
106	Sulekha	Srivastava	5-6-2006	10000

Table: COURSES

C23	104	Computer Security	8000
C24	106	Human Biology	15000
C25	102	Computer Network	20000
C26	105	Visual Basic	6000

i) To display complete details (from both the tables) of those Faculties whose salary is less than 12000.
ii) To display the details of courses whose fees is in the range of 20000 to 50000 (both values included).
iii) To increase the fees of all courses by 500 which have "Computer" in their Course name
iv) (A) To display names (FName and LName) of faculty taking System Design.

OR

(B) To display the Cartesian Product of these two tables

CS 2025 Sahodya Paper

1 Marks

SET 1

1. The ABC Company is considering to maintain their salespersons records using SQL to store data. As a database administrator, Alia created the table Salesperson with 6 columns and also entered the data of 5 salespersons. After a few days she added two or more columns and 4 more salespersons details. What is the degree and cardinality of the table Salesperson?
2. Which SQL command is used to modify data records in a relation?
3. Select the correct statement with reference to RDBMS:
   a) NULL can be a value in a primary key column.
   b) (Empty string) can be a value in a primary key column.
   c) A table with a primary key column cannot have an alternate key.
   d) A table with primary key column must have an alternate key.

4. Which among the following is not the characteristic of a relation?
   a) The ordering of attributes is important.
   b) The value in each tuple is an atomic value.
   c) If the value of an attribute in a tuple is not known or not applicable or not available, a special value called NULL is used to represent them.
   d) No two tuples of relation should be identical.
5. Getta wants to know the suage of NULL in MySQL. Help her choose in which of the following case NULL value cannot be assigned to the column Admission-No:
   a) When the Admission-No is zero.
   b) When the Admission-No is not known.
   c) When the Admission-No is not available.
   d) When the Admission-No is not applicable.

SET 2

1. Identify the SQL command to delete a relation from a relational database.
   a. DROP TABLE　　b. REMOVE TABLE
   c. DELETE TABLE　　d. ERASE TABLE
2. Sonal needs to display name of teachers, who have "O" as the third character in their name. She wrote the following query.
   SELECT NAME FROM TEACHER WHERE NAME = "%O%";
   But the query isn't producing the result. Rewrite the correct query.
3. _______ is the table constraint used to stop null values to be entered in the field.
   a. Unique　　b. Not NULL
   c. default　　d. Foreign Key
4. Which of the following types of table constraints will prevent the printing of duplicate values after the Select statement fetches data from table?
   a. Unique　　b. Distinct
   c. Primary Key　　d. NULL

2 Marks

SET 1

1(a) Reema is creating a table in a database, where none of the items are NULL. She wants to identify every record uniquely. What constraint is best suited here?

OR

(b) This constraint can only be applied to one column or group of columns together. It cannot be applied to one column individually for a table. Which constraint is referred here?

2 (a) Write an SQL command to add an attribute quantity of integer type to an existing table, named product.

OR

(b) Rita has created a table named pharma with attributes medicine_id, name and price. She forgot to set primary key constraint. Help her to do the given tasks.

SET 2

Explain the use of 'Foreign Key' in a Relational Database. Give an example to support your answer.

OR

What do you mean by aliasing with respect to SQL? Give a suitable example for the same.

4 Marks

SET 1

1. Consider the table Persons:

Pid	Surname	Firstname	Gender	City	Pincode	Salary
1	Sharma	Geeta	F	Hassan	182141	50000
2	Singh	Surinder	M	Bangalore	181165	42000
3	Jacob	Peter	M	Manipal	184232	45000
4	Alvis	Thomas	M	Manipal	184232	70000
5	Mohan	Garima	M	Bangalore	181165	74000
6	Azmi	Simi	F	Hassan	182141	68000
7	Kaur	Manpreet	F	Bangalore	181165	66000

Note: The table contains many more records than shown here.

a) Write the following SQL Queries:
   i) To display the total salary city wise excluding the cities with count more than two.
   ii) To display all the first names of male gender in descending order of their salary.
   iii) To display the details of all the persons whose city name has letter 'l' in it.
   iv) Display the increased salary of every person by 5%.

b) Write the output of the queries given below:
   i) SELECT DISTINCY City FROM Persons;
   ii) SELECT * FROM Persons WHERE City NOT IN ('Bangalore', 'Manipal');
   iii) SELECT Surname, Firstname, Pincode FROM Persons WHERE Salary BETWEEN 40000 and 5000;
   iv) SELECT MAX(Salary), MIN(Salary) FROM Persons;

2. Consider the following tables Train and Passenger:

Table: Train

TID	TNAME	BOARDING	DESTINATION	COST
305	VandeBharat	New Delhi	Katra	2500
206	Shatabdi	Mumbai	Delhi	3500
307	Superfast	Lucknow	Chennai	4500
208	HolidayExpress	Goa	Delhi	2000

Table: Passenger

PID	TID	NAME	DOT
P108	307	Gagandeep Singh	2022-10-14
P209	206	Rahat Ali	2023-01-01
P110	305	Cristopher	2022-12-10
P677	208	Diksha Luther	2019-09-11
P555	307	Nira Kadam	2020-11-06
P211	208	Ritam Ahuja	2021-12-17

Write the SQL Queries for the following:

a) To display the count of passengers travelling by each train.
b) To increase the cost by 15% for all the trains where destination is Delhi.

c) To display Tname, Name and Boarding of all passengers.
d) To delete the details of those passengers who travelled in 2019.

OR

To display the name of passengers boarding from Goa.

SET 2

1. Consider the following table DOCTOR as given below:

D_Id	D_Name	D_Dept	Gender	Experience
101	Joseph	ENT	Male	10
104	Gupta	Medicine	Male	12
106	Suman	Ortho	Female	7
111	Haneef	ENT	Male	12
123	Deepti	NULL	Female	6
123	Veena	Skin	Female	12

a) Write the SQL queries for the following:
   i) To display all the details of female doctors whose experience is more than 10.
   ii) Display the names of male doctors in the descending order of their experience.
   iii) To change the department of the Doctor 106 from ortho to ENT.
   iv) To display the number of doctors who is not assigned to any department. (NULL in the department)
   OR

b) Write the output for the following queries:
   i) SELECT D_NAME, GENDER from DOCTOR WHERE EXPERIENCE = 6 AND GENDER != 'MALE';
   ii) SELECT GENDER,MIN(EXPERIENCE) FROM DOCTOR GROUP BY GENDER;
   iii) SELECT COUNT(DISTINCT D_DEPT) FROM DOCTOR;
   iv) SELECT D_NAME, EXPERIENCE FROM DOCTOR WHERE NOT D_DEPT IN ('ORTHO','ENT');

2. Consider the following tables PRODUCT and CLIENT given below:

Table: Product

Pr-Id	Pr-Name	Manufacturer	Price	Qty
BS101	Bath Soap	Fiama	45.00	25
BS120	Bath Soap	Santoor	36.00	10
BS145	Bath Soap	Dove	38.00	20
FW422	Face Wash	Dettol	66.00	10
SP210	Shampoo	Sunsilk	320.00	10
SP235	Shampoo	Dove	455.00	15
TB310	Toothbrush	Colgate	48.00	15

Table: Client

C-Id	C-Name	City	Pr-Id
1	Dream Mart	Bangalore	BS101
2	Shoprix	Dehi	TB310
3	Big Bazar	Delhi	SP235
4	Metro	Chennai	FW422

i) Query to display the PR_ID, PR-Name and C-Name from product and client table whose manufactures are either FIAMA OR COLGATE.
ii) Query to display the details of those PRODUCTS whose client city is DELHI.
iii) Query to display the manufacturer and Price x Qty as Total for the manufacturer's whose name ends with 'E'.
iv) Query to add a column called Discount (Int Data type) in the Product table.

CS SUPPLEMENTARY PAPER 2025

1 Marks

1. In a particular examination, there are 50 candidates with roll numbers from 997601 to 997650. The data of these candidates is stored in a table in a database. What is the domain of the column which stores the roll numbers of the candidates?
2. Write any two DDL commands.
3. Which of the following is not a valid relational operator used with WHERE clause in SQL?
   a) > b) <=
   c) => d) <>
4. What is the default format of date in MySQL?
   a) DD-MM-YYYY b) MM-DD-YYYY
   c) YYYY-MM-DD d) YYYY-DD-MM
5. Ginni has created a table, SCORES in MySQL to store runs scored by players in a cricket match. The table contains the following records:

Table: Scores

PLAYER	SCORE
Ranveer	50
Sukesh	35
Mirza	10
John	51
Murugan	70

Which of the following statements will give 10 as output?
   a) SELECT MAX(Score) FROM Scores;
   b) SELECT MIN(Score) FROM Scores;
   c) SELECT SUM(Score) FROM Scores;
   d) SELECT AVG(Score) FROM Scores;

2 Marks

1. a) Write any one difference between CHAR and VARCHAR data types in MySQL.

   OR

   (b) Write one difference between Primary key and Unique constraint.
2. a) Write an SQL command to remove a column named ADDRESS, from a table named CUSTOMER.

   OR

   (b) Write an SQL command to add a column named ADDRESS, of type VARCHAR (20) in a table named CUSTOMER.

4 Marks

1. Consider the table STAFF given below:

Table : STAFF

STAFF_ID	STAFF_NAME	SALARY	DEPARTMENT	DESIGNATION
S101	SUNITA	26000	MATHS	TGT
S201	SUNIL	80000	COMMERCE	PGT
S301	NEHA	35000	SCIENCE	TGT
S102	MANJEET	25000	MATHS	TGT
S202	MANNAN	45000	COMPUTER	TGT

(a) Write the suitable SQL queries to perform the following tasks:

i) To display the average salary of each department.
ii) To insert the following record in the table,
STAFF_ID: S333, STAFF_NAME: GURMEET, SALARY: 15000, DEPARTMENT: ADMIN, DESIGNATION: CLERK
iii) To display the unique designations from the table.
iv) To display all the details of the staff whose name is of four letters.

OR

(b) Write the output for the queries given below:

i) SELECT STAFF_NAME FROM STAFF WHERE SALARY BETWEEN 25000 AND 30000;
ii) SELECT * FROM STAFF WHERE DEPARTMENT = "MATHS" AND SALARY > 25000;
iii) SELECT STAFF_NAME, STAFF_ID FROM STAFF WHERE DEPARTMENT LIKE "%S";
iv) SELECT MAX(SALARY)FROM STAFF;

2. Assume that you are working for ABC Corporation (ABCC). ABCC allots contracts to different contractors for some of its works. The data of Contracts and Contractors are kept in the tables Work and Contractor respectively. Following are a few records from these two tables of ABCC's database.

Table : **Work**

W_ID	C_ID	W_Name	W_Amt
P0001	C_01	Painting	20000
E0001	C_01	Electrical	50000
D0001	C_02	Dumping	10000

Table : **Contractor**

C_ID	C_Name	Phone	email
C_01	M. Khan & Sons	1232311	MK@xyz.com
C_02	Acharya Pvt. Ltd.	2323311	APL@xyz.com
C_03	Charu Corp.	NULL	CCP@pqr.xyz

Note: The tables contain many more records than shown here.
As an employee of ABCC, you are required to write the SQL queries for the following:

i) To display all the records from the Work table in alphabetical order of W-Name.
ii) To display the names of contractors where W-Amt is more than 15000.
iii) To display the structure of Work table.
iv) (a) To count total number of records, present in Work table.

OR

(b) To delete the records of contractors whose phone number is not known.

CS MAIN PAPER 2026

1 Marks

1. A table has two candidate keys, one of which is chosen as the primary key. How many alternate keys does this table have?
   a) 0 b) 1
   c) 2 d) 3
2. Which of the following SQL command can change the degree of the existing relation?
   a) DROP TABLE
   b) ALTER TABLE
   c) UPDATE…SET
   d) DELETE
3. What will be the output of the query?
   SELECT MACHINE_ID, MACHINE_NAME FROM INVENTORY WHERE QUANTITY <= 100;
   a) All columns of INVENTORY table with quantity greater than 100.
   b) ID and name of machines with quantity less than 100 from INVENTORY table.
   c) All columns of INVENTORY table with quantity greater than or equal to 100.
   d) ID and name of machines with quantity less than or equal to 100 from INVENTORY table.
4. A relation in MySQL database consists of 2 tuples and 3 attributes. If 2 attributes are deleted and 4 tuples are added, what will be the cardinality of the relation?
   a) 4 b) 6
   c) 5 d) 7
5. Which aggregate function in SQL returns the smallest value from a column in a table?
   a) MIN() b) SMALL()
   c) MAX() d) LOWER()

2 Marks

1. Ms. Zoya is a Production Manager in a factory which packages mineral water. She decides to create a table in a database to keep track of the stock present in the factory. Each record of the table will have the following fields:
   W-Code – Code of the item (type – CHAR(5))
   W-Description – Description of the item (type – VARCHAR(20))
   B-Qty – Balance quantity of the item (type – INTEGER)
   U-Price – Unit Price of the item (type – FLOAT)
   The name of the table is W_STOCK.
   i) (a) Write an SQL command to create the above table (W-Code should be the primary key).

   OR

   (b) Can U-Price be the primary key of the above table? Justify your answer.

   ii) (a) Assuming that the table W_STOCK is already created, write an SQL command to add an attribute E-Date (of DATE type) to the table.

   OR

   (b) Assuming that the table W_STOCK is already created, write an SQL command to remove the column B-Qty from the table.

4 Marks

1. Abhishek has created a table, named STOCK, with a set of records to maintain the data of packaged milk in his shop. After creating the table, he entered the data and the table looked as follows:

Code	Type	Volume	Qty	Price
AF0.5	F	0.5	300	38.00
MF0.5	F	0.5	250	36.50
MT1.0	T	1.0	150	64.00
AT1.0	T	1.0	100	66.0
PD1.0	D	1.0	50	52.00
PT0.5	T	0.5	78	30.00

   a) Based on the data given above, write the SQL queries for the following tasks:
      i) To display Type and the maximum Price for each Type of milk.
      ii) For each record, increase the Price by 0.5 where Type is 'F'.
      iii) To display the total value of the stock (total of Qty × Price).
      iv) To display the details of all records where Code starts with 'A'.

   OR

   b) Considering the table STOCK as given above, write the output on execution of the following queries:
      i) SELECT Volume, Qty, Price FROM STOCK WHERE Type IN ('F','D');
      ii) SELECT Code, Qty FROM STOCK WHERE Price BETWEEN 30 AND 50;
      iii) SELECT DISTINCT Type FROM STOCK;
      iv) SELECT Volume, count(*) FROM STOCK GROUP BY Volume;

2. Assume that you are the Manager of the Loans department of a Finance House. To keep track of the loans you have created two tables: CUSTOMERS and LOANS.
The sample data in these tables is given below:

TABLE: CUSTOMERS

C_ID	C_NAME	PHONE
00001	Raj Malhotra	1234567890
00003	David Xavier	3456789012
00004	Damini Iyer	3156789012
00008	Abdul	2345678901

TABLE: LOANS

Sno	C_Id	Loan_Amt	L_Date	Terms	ROI
1	00003	200000	2025-12-06	60	7.80
2	00008	2500000	2023-08-09	60	9.00
3	00001	500000	2025-08-13	48	6.00
4	00003	300000	2026-12-07	36	8.00
5	00004	600000	2026-12-07	60	6.00

Note: The tables may contain more records than shown here.
The management of the Finance House needs certain reports from you. Write the queries to extract the following data to create the reports:

i) Number of records from LOANS table where Rate of Interest (ROI) is above 7.0.
ii) Names of the customers whose loan amount (L-Amt) is above 1000000.
iii) C_ID, C-Name and Terms of all those records where Loan Date (L-Date) is after 31st December, 2024.
iv) (a) Details of all the loans in the descending order of ROI.

OR

(b) C_ID and average term for each C_ID from the LOANS table.

CS SAMPLE PAPER 2026

1 Marks

1. Consider the given SQL Query:
SELECT department, COUNT(*) FROM employees HAVING COUNT(*) > 5 GROUP BY department;
Saanvi is executing the query but not getting the correct output. Write the correction.
2. Which SQL command can change the cardinality of an existing relation?
a) Insert b) Delete
c) Both a) & b) d) Drop
3. In SQL, a relation consists of 5 columns and 6 rows. If 2 columns and 3 rows are added to the existing relation, what will be the updated degree of a relation?
a) Degree: 7 b) Degree: 8
c) Degree: 9 d) Degree: 6
4. Which SQL command is used to remove a column from a table in MySQL?
a) UPDATE b) ALTER
c) DROP d) DELETE

2 Marks

1. Write suitable commands to do the following in MySQL.
   a) View the table structure.
   b) Create a database named SQP

OR

Differentiate between drop and delete query in SQL with a suitable example.

4 Marks

1. Consider the table SALES as given below:

Sales-id	Customer-name	Product	Quantity-sold	Price
S001	John Doe	Laptop	5	50000
S002	Jane Smith	Smartphone	10	30000
S003	Michael Lee	Tablet	3	15000
S004	Sarah Brown	Headphones	7	2000
S005	Emily Davis	Smartwatch	8	8000
S006	David	Smartwatch	3	16000
S007	Mark	Tablet	5	34000

A. Write the following queries:
   i) To display the total quantity sold for each product whose total quantity sold exceeds 12.
   ii) To display the records of SALES table sorted by Product name in descending order.
   iii) To display the distinct Product names from the SALES table.
   iv) To display the records of customers whose names end with the letter 'e'.

OR

B. Predict the output of the following:
   i) SELECT * FROM Sales where product='Tablet';
   ii) SELECT sales-id, customer-name FROM Sales WHERE product LIKE 'S%';
   iii) SELECT COUNT(*) FROM Sales WHERE product in ('Laptop', 'Tablet');
   iv) SELECT AVG(price) FROM Sales where product='Tablet';

2. Pranav is managing a Travel Database and needs to access certain information from the Hotels and Bookings tables for an upcoming tourism survey. Help him extract the required information by writing the appropriate SQL queries as per the tasks mentioned below:

Table: Hotels

H-ID	Hotel-Name	City	Star-Rating
1	Hotel1	Delhi	5
2	Hotel2	Mumbai	5
3	Hotel3	Hyderabad	4
4	Hotel4	Bengaluru	5
5	Hotel5	Chennai	4
6	Hotel6	Kolkata	4

Table: Bookings

B-ID	H-ID	Customer-Name	Check-In	Check-Out
1	1	Jiya	2024-12-01	2024-12-05
2	2	Priya	2024-12-03	2024-12-07
3	3	Alicia	2024-12-01	2024-12-06
4	4	Bhavik	2024-12-02	2024-12-03
5	5	Charu	2024-12-01	2024-12-02
6	6	Esha	2024-12-04	2024-12-08
7	6	Dia	2024-12-02	2024-12-06
8	4	Sonia	2024-12-04	2024-12-08

i) To display a list of customer names who have bookings in any hotel of 'Delhi' city.
ii) To display the booking details for customers who have booked hotels in 'Mumbai', 'Chennai', or 'Kolkata'.
iii) To delete all bookings where the check-in date is before 2024-12-03.
iv) A. To display the Cartesian Product of the two tables.

OR

B. To display the customer's name along with their booked hotel's name.

CS 2026 Sahodya Paper

1 Marks

1. Consider the given SQL query:
   SELECT department, AVG(Salary) FROM employees WHERE salary>50000 GROUP BY department HAVING COUNT(*)>=3 ORDER BY AVG(salary);
   Identify the correct sequence of execution for this query.
   a) WHERE->SELECT->GROUP BY-> HAVING->ORDER BY
   b) GROUP BY->WHERE->HAVING->SELECT->ORDER BY
   c) SELECT->WHERE->GROUP BY->HAVING->ORDER BY
   d) WHERE->GROUP BY->HAVING->SELECT->ORDER BY
2. Which combination of SQL commands will both change the degree and cardinality of a relation?
   a) ALTER TABLE and UPDATE
   b) UPDATE and DELETE
   c) ALTER TABLE and INSERT
   d) DROP and CREATE
3. A relation R1 has degree 6 and cardinality 10. Another relation R2 has degree 4 and cardinality 8. If a natural join is performed on these relations with two common attributes resulting in 12 rows, what will the degree of the resultant relation be?
   a) 8 b) 10 c) 6 d) 4
4. Complete the following SQL statement to display the vehicles registered in any district of Uttar Pradesh, using any of the option given below:
   SELECT makeyear, vehname FROM vehicle WHERE v-no ___________;
   a) ='UP' b) like 'UP' c) like 'UP%' d) like 'UP__'

2 Marks

1 (a) Write suitable MySQL commands for the following:

i) Change the datatype of a column name phone-no in table Supplier from varchar(15) to int.

ii) Remove all records from a table named Student but retain the table structure.

OR

(b) Differentiate between where clause and having clause in SQL with suitable examples showing when each should be used.

4 Marks

1. Consider the table Employees as given below:

Emp-Id	Name	Department	Designation	Salary	Join-Date	City
E101	Rajesh Kumar	Sales	Manager	85000	2020-03-15	Mumbai
E102	Anita Sharma	IT	Developer	75000	2021-06-20	Delhi
E103	Vikram Singh	Sales	Executive	45000	2022-01-10	Mumbai
E104	Priya Patel	HR	Manager	80000	2019-08-25	Bangalore
E105	Amit Verma	IT	Developer	72000	2021-11-30	Delhi
E106	Sneha Reddy	IT	Lead	95000	2018-05-12	Bangalore
E107	Karan Mehta	Sales	Manager	88000	2020-09-18	Mumbai
E108	Divya Nair	HR	Executive	48000	2023-02-14	Bangalore

a) Write SQL queries for the following:

i) Display the Name, Department and Salary of employees whose salary is between 60000 and 90000 and working in the HR or Finance department.

ii) Display each department and number of employees in it. Show only those department having more than 2 employees.

iii) Display name, city and salary increased by 8%(display only, don't update) for employees whose city name starts with M.

iv) Lis Emp-id, Name and Salary of employees who joined after 2020 and sort salary in descending order.

OR

b) Predict the output of the following SQL queries (Assume suitable data in Employee table):

i) SELECT Department, COUNT(*) AS Empcount FROM Employees GROUP BY Department HAVING COUNT(*)>2;

ii) SELECT Name, Department, SALARY*0.010 AS Bonus FROM Employees WHERE Department='Sales';

iii) SELECT City, MIN(Salary) AS Minsal, MAX(Salary) AS Maxsal FROM Employees GROUP BY City;

iv) SELECT Name, Designation, Salary FROM Employees WHERE Salary>=80000 ORDER BY Salary DESC;

2. Meera is managing a college database with the following tables:

Table: Courses

Course-Id	Course-Name	Department	Credits	Instructor
C101	Data Structures	CS	4	Dr. Smith
C102	Digital Logic	CS	3	Dr. Jones
C103	Thermodynamics	ME	4	Dr. Kumar
C104	Database Systems	Cs	4	Dr. Smith
C105	CAD	ME	3	Dr. Sharma

Table: Enrollments

Enrollment-ID	Student-Name	Course-Id	Semester	Grade
E001	Amit	C101	III	A
E002	Priya	C101	III	B
E003	Raj	C102	II	A
E004	Amit	C104	I	A
E005	Sara	C103	II	C
E006	Priya	C104	III	A
E007	Raj	C101	IV	C

Write SQL queries for the following:

i) Write a query to change the Credits to 5 for the course whose Course id is C105.

ii) Display Student name, Course name, Grade and Semester of all students enrolled in Semester III, using a join between courses and enrollment tables.

iii) Write a query to delete all records from enrollments where grade is c.

iv) Write a query to display student name, course name and department for all students who are enrolled in any course where the instructor's name ends with 'smith'.

OR

Display department, total number of courses in that department, and the average credits per course. Show only those departments where the average credits are greater than 3.